

Multivariate Meta-Analysis for Longevity and Reproduction

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This documents the exploratory multivariate meta-analysis for the effects of temperature on the longevity and reproduction of species.

Multivariate meta-analysis and meta-regression

Reproductive traits were more strongly impacted by temperature changes than longevity when controlling for temperature (contrast = -1.04, 95% CI: -1.31 to -0.76, $df = 396$, p -value = < 0.0001). At 25°C, longevity was reduced by -0.22 SD units relative to the control (95% CI: -0.41 to -0.04, $df = 396$, p -value = 0.02). In contrast, reproduction was significantly reduced by -1.26 SD units relative to the control (95% CI: -1.52 to -1, $df = 396$, p -value = < 0.0001). However, there was high heterogeneity in both traits with prediction intervals spanning -6.35 to 4.12 (Figure 1).

Mean effects changed in complicated ways with changes in temperature, with significant non-linear patterns that depended on the effect outcome (Table 1), with effect magnitude and direction changing non-linearly with temperature increases. Interestingly, studies that observed larger changes in reproduction also observed correlated changes in longevity (between study correlation = 0.37), however, at the within study-level there was a much weaker correlation (within-study correlation = 0.17, Figure 2).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.0903	-0.3092	0.1286	392	0.42
Reproduction (mean)	-0.4837	-0.7943	-0.1732	392	< 0.01
Linear Treatment Temperature (centered) - Longevity	-0.2581	-0.2875	-0.2288	1,576	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	-0.0660	-0.0978	-0.0342	1,576	< 0.0001
Quadratic Treatment Temperature (centered) - Longevity	-0.0022	-0.0042	-0.0002	1,576	0.03

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Quadratic Treatment Temperature (centered) - Reproduction	-0.0123	-0.0146	-0.0099	1,576	< 0.0001
Cubic Treatment Temperature (centered) - Longevity	0.0005	0.0003	0.0006	1,576	< 0.0001
Cubic Treatment Temperature (centered) - Reproduction	-0.0001	-0.0002	0.0001	1,576	0.42

Table 1: Non-linear effects of temperature on mean effect size for longevity and reproduction

Sensitivity Analysis

In this analysis we remove any effect sizes above 10 or below -10. When removing the 10% largest and smallest effect sizes, the following effect sizes are removed. We found very little significant differences upon removing outlying effect size studies (Table 2).

[1] TEMP559_A.-10 TEMP1964A.5 TEMP1964A.7 TEMP2281A.10 TEMP546_A.10
[6] TEMP559_A.15
21976 Levels: HUM109A.-26 HUM148A.-26 HUM22B.-26 HUM256A.-26 ... TEMP995A.21

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.0286	-0.2051	0.1480	388	0.75
Reproduction (mean)	-0.4147	-0.6512	-0.1781	388	< 0.001
Linear Treatment Temperature (centered) - Longevity	-0.2142	-0.2347	-0.1937	1,532	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	-0.0414	-0.0665	-0.0164	1,532	< 0.01
Quadratic Treatment Temperature (centered) - Longevity	-0.0026	-0.0040	-0.0011	1,532	< 0.001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0090	-0.0109	-0.0070	1,532	< 0.0001
Cubic Treatment Temperature (centered) - Longevity	0.0004	0.0003	0.0005	1,532	< 0.0001
Cubic Treatment Temperature (centered) - Reproduction	0.0000	-0.0002	0.0001	1,532	0.6

Table 2: Non-linear effects of temperature on mean effect size for longevity and reproduction after removing outlying effect sizes.

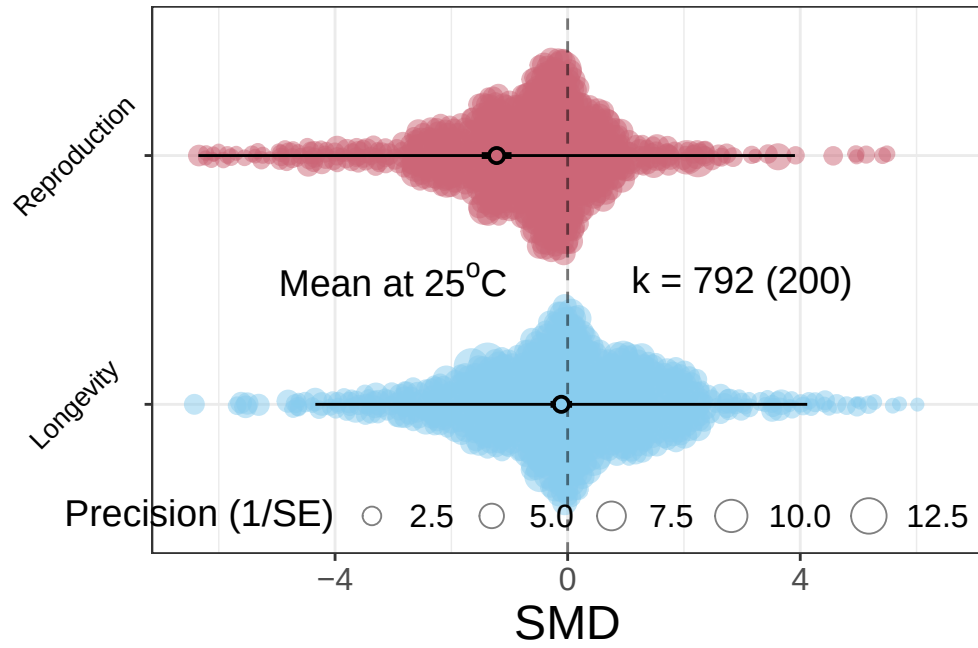


Figure 1: Orchard plot with the overall meta-analytic mean (central dots) for reproduction and longevity traits. Thick bars are the 95 confidence intervals and thin bars are the 95 prediction intervals. Data are weighted by their precision (inverse sampling error). Plot is truncated for ease of visualisation. k is the number of effect sizes and n the number of studies. Given we limited our analysis to studies with both reproduction and longevity the samples sizes are the same for both trait types.

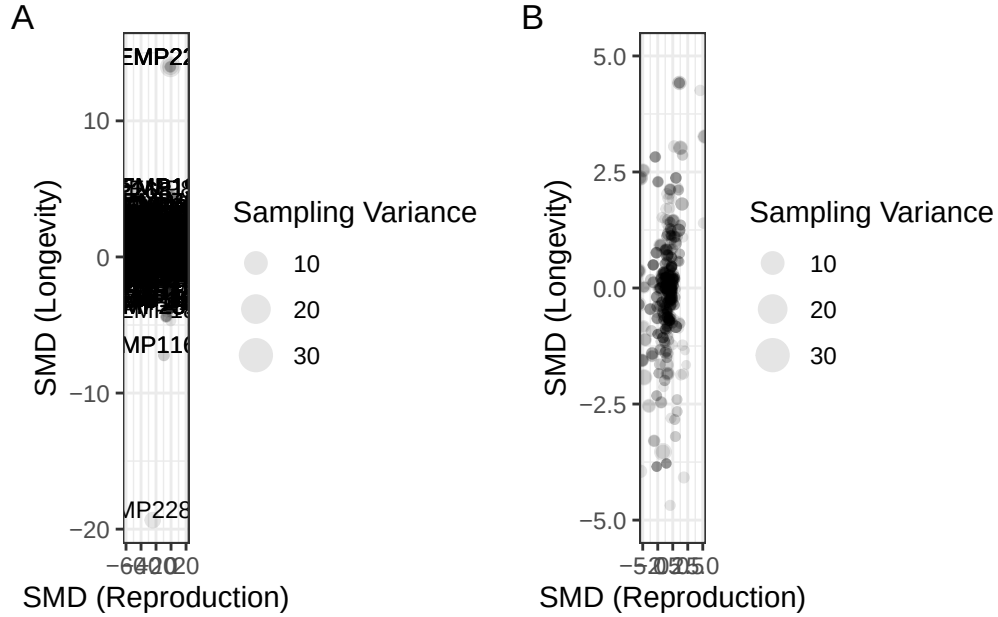


Figure 2

Splitting analysis: warm versus cool

In this analysis we split the dataset into studies warmer and cooler than the reference temperature of 25 degrees C. Here we explored non-linear effects up to degree 4. We found that there was a significant and negative linear effect of temperature on both reproduction and adult lifespan at warmer temperatures (-0.17 and -0.38 respectively, (Table 3). However, reproductive traits were more strongly impacted by increased temperature changes than adult lifespan (contrast = -0.21, 95% CI: -0.26 to -0.16, $df = 786$, t -value = -8.59, p -value = < 0.0001).

For treatment temperatures that were cooler than the reference we found a significant and negative quadratic relationship between temperature and lifespan (est. -0.01, 95% CI: -0.02 to -0.01). Adult lifespan is estimated to increase with cooler temperatures with the optimum predicted lifespan to be approximately 12°C cooler than the reference temperature. In contrast, we found a significant linear relationship between temperature and reproduction (est. 0.15, 95% CI: 0.08 to 0.22). However, we found no significant quadratic relationship between temperature and reproduction (see Table 4 for estimates). Figure 3 shows the predicted relationship between temperature and reproduction and adult lifespan for observed treatment temperatures warmer (plot A) and cooler (plot B) than their reference temperature.

Interestingly, Figure 4 shows that studies which observed larger changes in reproduction also observed correlated changes in longevity (between study correlation = 0.65, 95% CI: 0.52 to

0.75), however, at the within study-level there was a much weaker correlation (within-study correlation = 0.5, 95% CI: 0.38 to 0.61).

We found that studies that observed larger changes in reproduction also observed correlated changes in longevity (between study correlation = -0.15, 95% CI: -0.31 to 0.09), however, at the within study-level there was a much weaker correlation (within-study correlation = -0.23, 95% CI: -0.36 to -0.09)

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.53	-0.89	-0.18	344	< 0.01
Reproduction (mean)	0.54	0.01	1.08	344	0.05
Linear Treatment Temperature (centered) - Longevity	-0.17	-0.21	-0.13	786	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	-0.38	-0.44	-0.32	786	< 0.0001

Table 3: Linear effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are warmer than reference temperatures.

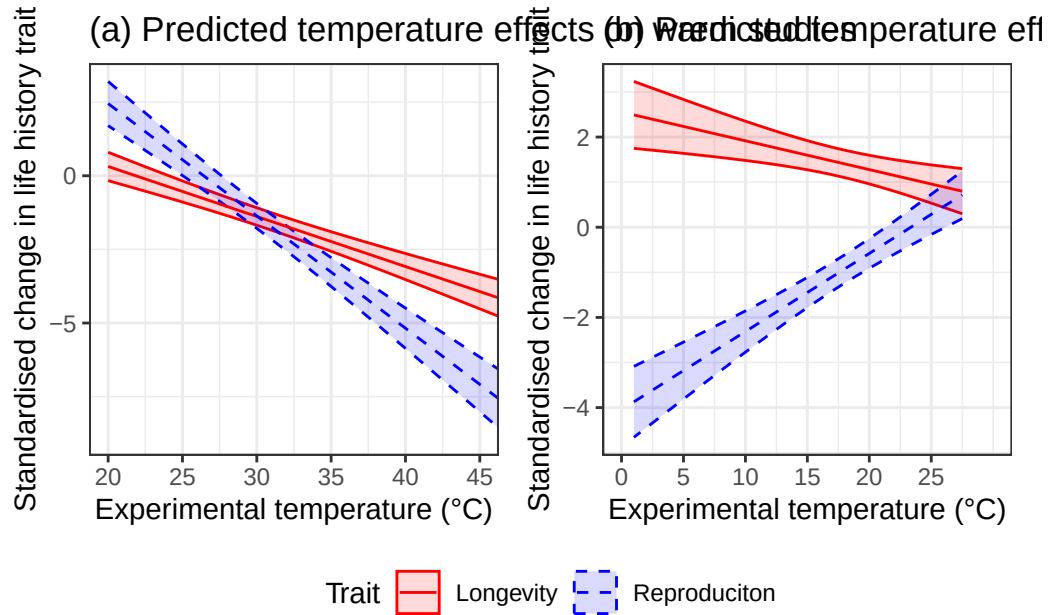


Figure 3

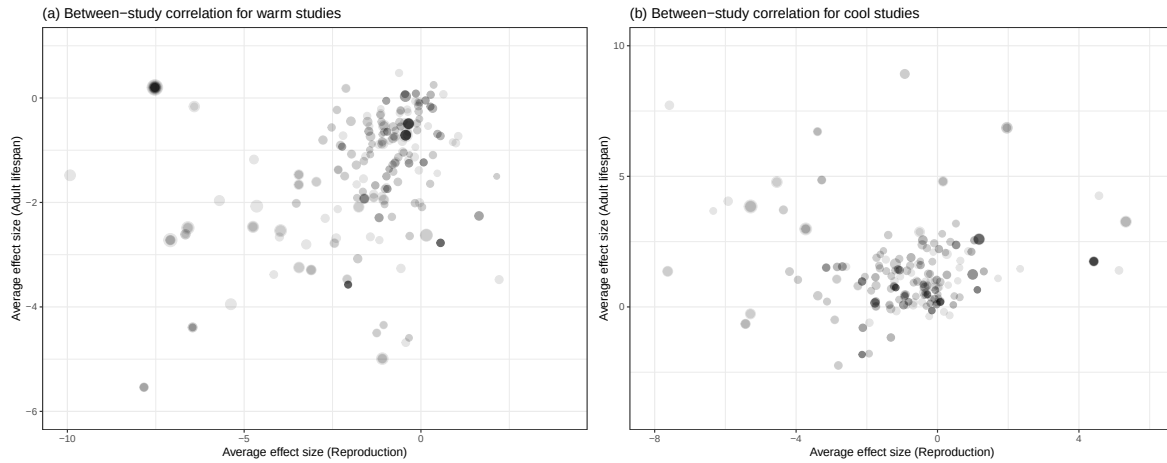


Figure 4

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.0972	-0.4646	0.6590	320	0.73
Reproduction (mean)	0.4945	-0.1196	1.1087	320	0.11
Linear Treatment Temperature (centered) - Longevity	-0.3271	-0.3955	-0.2586	790	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1484	0.0754	0.2214	790	< 0.001
Quadratic Treatment Temperature (centered) - Longevity	-0.0146	-0.0191	-0.0101	790	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0003	-0.0052	0.0046	790	0.91

Table 4: Quadratic effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures.

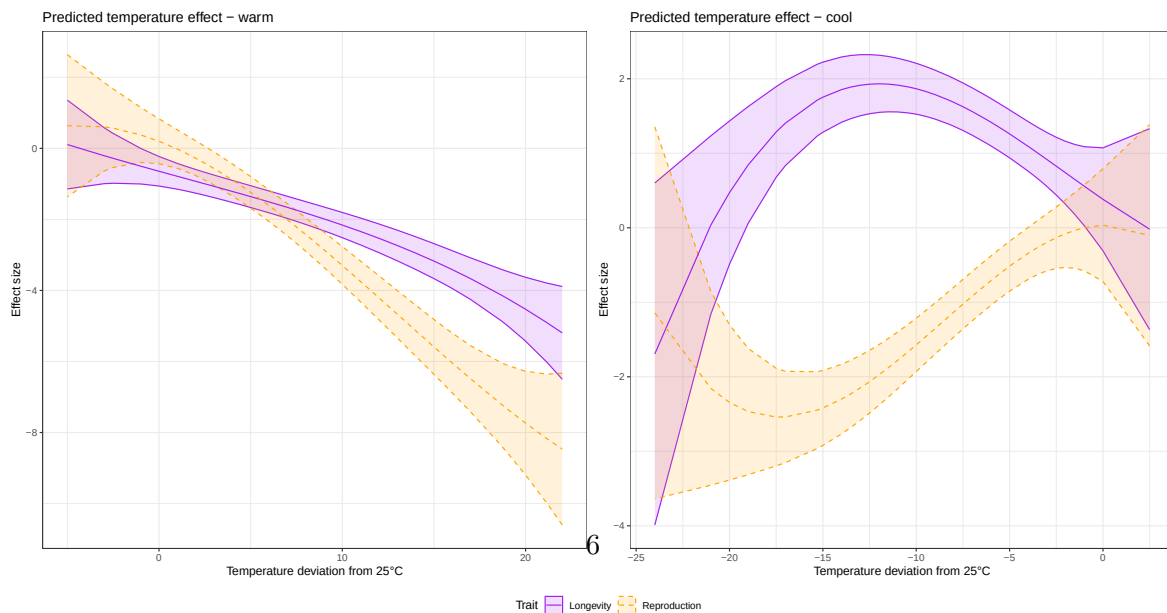


Figure 5

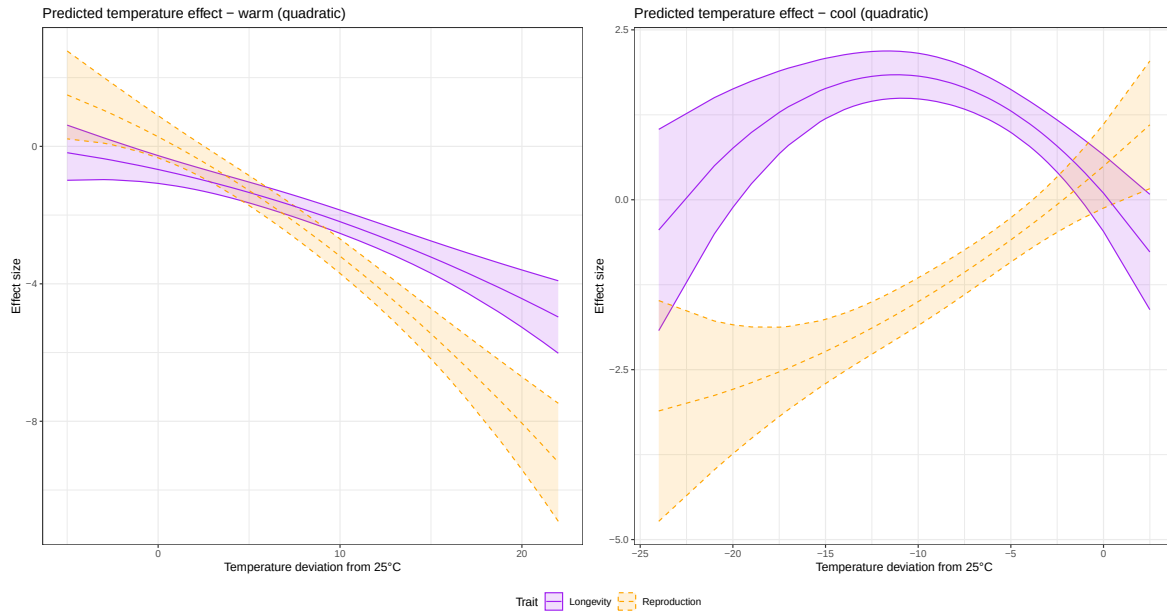


Figure 6

Table 5: quadratic effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures (sensitivity analysis with 10% outliers removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.3407	-0.1321	0.813	317	0.16
Reproduction (mean)	0.3858	-0.1916	0.963	317	0.19
Linear Treatment Temperature (centered) - Longevity	-0.2892	-0.3448	-0.234	773	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1403	0.0729	0.208	773	< 0.0001
Quadratic Treatment Temperature (centered) - Longevity	-0.0139	-0.0177	-0.010	773	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	0.0004	-0.0041	0.005	773	0.85

We found qualitatively similar correlations, after removing outliers (between-study correlation = -0.03, 95% CI: -0.24 to 0.19 within-study correlation = -0.15, 95% CI: -0.29 to 0)

Additional analyses

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.57	-0.991	-0.145	153	< 0.01
Reproduction (mean)	-0.12	-0.196	-0.045	705	< 0.01
Linear Treatment Temperature (centered) - Longevity	-0.43	-0.807	-0.059	312	0.02
Linear Treatment Temperature (centered) - Reproduction	0.62	0.031	1.217	312	0.04
Quadratic Treatment Temperature (centered) - Longevity	-0.17	-0.207	-0.138	706	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.39	-0.449	-0.324	706	< 0.0001

Table 6: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are warmer than reference temperatures (with Aquatic species removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.2861	-0.1055	0.6777	144	0.15
Reproduction (mean)	-0.2841	-0.3775	-0.1908	713	< 0.0001
Linear Treatment Temperature (centered) - Longevity	-0.3296	-0.3996	-0.2595	714	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1551	0.0775	0.2327	714	< 0.001
Quadratic Treatment Temperature (centered) - Longevity	-0.0148	-0.0193	-0.0103	714	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0002	-0.0053	0.0049	714	0.94

Table 7: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures (with Aquatic species removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.47	-0.837	-0.11	123	0.01
Reproduction (mean)	1.08	0.516	1.64	250	< 0.001
Linear Treatment Temperature (centered) - Longevity	-0.47	-0.835	-0.11	250	0.01
Linear Treatment Temperature (centered) - Reproduction	0.61	-0.047	1.26	250	0.07
Quadratic Treatment Temperature (centered) - Longevity	-0.16	-0.197	-0.12	580	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.41	-0.478	-0.34	580	< 0.0001

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
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Table 8: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are warmer than reference temperatures (with assumed rearing temp studies removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.2601	-0.139	0.6588	108	0.2
Reproduction (mean)	-0.2608	-0.351	-0.1703	549	< 0.0001
Linear Treatment Temperature (centered) - Longevity	-0.3019	-0.368	-0.2360	550	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1625	0.065	0.2604	550	< 0.01
Quadratic Treatment Temperature (centered) - Longevity	-0.0134	-0.018	-0.0091	550	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0004	-0.007	0.0062	550	0.9

Table 9: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures (with assumed rearing temp studies removed).